

Centrale Supélec

Master BDMA - Decision Modeling

**Decision Models for the Nutri-Score Food Label:
Integrating Nutritional and Environmental Criteria**

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January 2026

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Abstract

This report presents a comprehensive study on elaborating decision models for the Nutri-Score food labeling system, with an extension to incorporate environmental impact through the Green-Score criterion. Using a dataset of 2,925 food products from Open Food Facts, we implement and compare three distinct approaches: the official Nutri-Score algorithm, ELECTRE TRI (a non-compensatory Multi-Criteria Decision Analysis method), the Weighted Sum Model (a compensatory approach), and machine learning techniques enhanced with SHAP explainability analysis.

Our results demonstrate that incorporating environmental criteria fundamentally changes product classifications, with approximately 80% of products receiving different ratings compared to nutrition-only assessments. The ELECTRE TRI model produces more conservative classifications due to its non-compensatory nature, while the Weighted Sum Model yields more balanced distributions. Machine learning models, particularly when combined with SHAP analysis, provide valuable insights into feature importance and validate our manually assigned criteria weights. We recommend the Weighted Sum Model with SHAP-validated feature importance for general consumer guidance applications, while acknowledging that different contexts may favor different approaches.

Keywords: Nutri-Score, Green-Score, ELECTRE TRI, Weighted Sum Model, MCDA, Machine Learning, SHAP, Food Labeling, Sustainability

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1. Introduction

Food labeling systems play a crucial role in informing consumer choices and promoting public health. In an era of increasing diet-related health issues including obesity, cardiovascular disease, and diabetes, accessible nutritional information has become essential for enabling informed purchasing decisions. The Nutri-Score system, developed in France and now adopted across several European countries, represents a significant advancement in simplified front-of-package nutrition labeling, converting complex nutritional data into an intuitive five-level color-coded scale from A (best) to E (worst).

However, modern food policy must address not only nutritional concerns but also environmental sustainability. The food system is responsible for approximately 25–30% of global greenhouse gas emissions, significant water usage, and substantial land-use impacts. Consumers increasingly express interest in making environmentally responsible choices, yet current labeling systems typically address nutrition and environment separately, creating a fragmented decision-making landscape.

This project addresses a fundamental gap in existing food labeling: the lack of integrated nutrition-environment assessment frameworks. Our objective is to develop and evaluate decision models that combine the seven nutritional criteria underlying the Nutri-Score with environmental impact data (Green-Score), creating a more comprehensive product evaluation system.

We employ three complementary methodological approaches. First, we implement and validate the official Nutri-Score algorithm. Second, we apply Multi-Criteria Decision Analysis (MCDA) methods, specifically ELECTRE TRI (a non-compensatory outranking method) and the Weighted Sum Model (a compensatory aggregation approach). Third, we explore machine learning techniques, enhanced with SHAP (SHapley Additive exPlanations) analysis to ensure interpretability.

The remainder of this report is structured as follows: Section 2 describes our database elaboration, Section 3 details the Nutri-Score implementation, Section 4 presents the ELECTRE TRI model, Section 5 covers the Weighted Sum Model, Section 6 explores machine learning approaches, Section 7 presents a comparison with another research group, and Section 8 discusses our findings and model recommendations.

2. Database Elaboration (Task 4.1)

2.1 Data Source and Collection

Our database was constructed from Open Food Facts, a collaborative, free, and open database of food products from around the world. We developed a Python script (get_data.py) to systematically query and filter products meeting our criteria. The final dataset comprises 2,925 food products, representing approximately fourteen times the minimum requirement of 200 products specified in the project guidelines.

Products were selected to ensure comprehensive coverage across both Nutri-Score and Green-Score classifications while maintaining diversity across food categories. The dataset includes 24 distinct food categories including cookies, bread, beverages, dairy products, snacks, plant-based foods, cereals, and desserts.

2.2 Database Schema

Each product record contains the following attributes: product_name, brands, categories, energy_100g (in kilojoules), saturated_fat_100g (in grams), sugars_100g (in grams), salt_100g (in grams), proteins_100g (in grams), fiber_100g (in grams), fvl_percent (fruits, vegetables, and legumes percentage), nutri_score_value, nutri_score_label, green_score_value (0–100 scale), and green_score_label.

2.3 Distribution Analysis

Table 1 presents the Nutri-Score distribution in our database. The minimum requirement specifies that each label should represent at least 15% of the dataset.

Label	Count	Percentage	Status
A	745	25.5%	✓ Satisfied
B	592	20.2%	✓ Satisfied
C	605	20.7%	✓ Satisfied
D	431	14.7%	Slightly below (15%)
E	552	18.9%	✓ Satisfied

Table 1: Nutri-Score distribution in the database

Table 2 presents the Green-Score distribution. The minimum requirement specifies that each label should represent at least 10% of the dataset.

Label	Count	Percentage	Status
A	890	30.4%	✓ Satisfied
B	891	30.5%	✓ Satisfied

C	529	18.1%	✓ Satisfied
D	397	13.6%	✓ Satisfied
E	218	7.5%	Slightly below (10%)

Table 2: Green-Score distribution in the database

We note slight shortfalls in Nutri-Score D (14.7% vs. required 15%) and Green-Score E (7.5% vs. required 10%). These minor deviations are attributable to the natural distribution of products in Open Food Facts: foods with the worst environmental scores (Green-E) are relatively rare in the database, as are products in the Nutri-Score D category which represents an intermediate poor rating. Attempts to artificially balance these categories would have required including lower-quality data entries or products with incomplete nutritional information, which we deemed counterproductive to our analytical objectives.

3. Nutri-Score Algorithm Implementation (Task 4.2)

3.1 Algorithm Description

The Nutri-Score algorithm computes a score based on two components:

- **Negative component (N):** Points assigned for nutrients to limit (energy, sugars, saturated fats, salt). Score ranges from 0 to 55 points.
- **Positive component (P):** Points assigned for nutrients to favor (proteins, fiber, fruits/vegetables/legumes). Score ranges from 0 to 17 points.

The final nutritional score is calculated as: *Nutritional Score* = $N - P$

A special rule applies: if $N \geq 11$ and fruits/vegetables/legumes content is $\leq 80\%$, protein points are not counted in the positive component.

3.2 Implementation Details

We implemented the Nutri-Score algorithm in Python following the official specifications from the ANSES (Agence nationale de sécurité sanitaire de l'alimentation) technical report. Our implementation includes:

- Point assignment tables for all seven criteria
- Logic for handling the special protein rule
- Classification into five categories (A through E) based on final score thresholds
- Validation against official examples from Open Food Facts

3.3 Graphical User Interface

We developed a web-based graphical user interface using Gradio, deployed on Hugging Face Spaces. The interface allows users to:

- Input the seven nutritional criteria values
- Receive the computed Nutri-Score label (A-E) and numerical score
- View a visual representation with color-coded output
- Access detailed breakdown of positive and negative component scores

The application is publicly accessible at: <https://huggingface.co/spaces/nutriScore-calculator>

3.4 Validation

We validated our implementation against the official Nutri-Score labels provided in Open Food Facts for all 2,925 products in our database. The implementation achieved 100% accuracy, confirming correct interpretation of the algorithm specifications.

4. ELECTRE TRI Model (Task 4.3)

4.1 Methodology

ELECTRE TRI (Élimination et Choix Traduisant la Réalité - TRI) is a non-compensatory multi-criteria decision analysis method designed for sorting problems. Unlike compensatory methods where poor performance on one criterion can be offset by excellent performance on another, ELECTRE TRI requires alternatives to meet minimum thresholds across multiple criteria simultaneously.

The method operates by comparing each alternative to limiting profiles that define boundaries between categories. Our implementation considers eight criteria: the seven Nutri-Score criteria (energy, sugars, saturated fats, salt, proteins, fiber, fruits/vegetables/legumes) plus the Green-Score as an eighth criterion.

4.2 Weight Assignment

We assigned weights to the eight criteria based on health and environmental policy priorities:

Criterion	Weight	Justification
Energy	0.10	Moderate importance; overconsumption contributes to obesity
Sugars	0.15	High importance; linked to diabetes and metabolic syndrome
Saturated Fats	0.12	High importance; cardiovascular disease risk factor
Salt	0.13	High importance; hypertension and cardiovascular risks
Proteins	0.10	Moderate importance; essential nutrient but widely available
Fiber	0.10	Moderate importance; digestive health and disease prevention
Fruits/Veg/Legumes	0.10	Moderate importance; micronutrients and health benefits
Green-Score	0.20	Highest importance;

		environmental sustainability priority
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Table 3: Criteria weights for ELECTRE TRI model

The Green-Score receives the highest weight (0.20) to reflect increasing policy emphasis on environmental sustainability. Among nutritional criteria, sugars and salt receive higher weights due to their strong associations with chronic diseases.

4.3 Limiting Profile Determination

We determined six limiting profiles (π_1 through π_6) defining the boundaries of five classes (A', B', C', D', E'). Our approach combined:

- **Statistical analysis:** Computing percentiles (20th, 40th, 60th, 80th) of criterion distributions to identify natural break points
- **Domain knowledge:** Adjusting thresholds based on nutritional and environmental standards
- **Validation:** Ensuring monotonicity (π_{k+1} strictly dominates π_k for all criteria)

The complete set of limiting profiles is provided in Table 4.

Profile	Energy	Sugars	Sat Fat	Salt	Protein	Fiber	FVL	Green
π_6 (upper)	3500	50	12	4.0	20	10	100	100
π_5 (A'/B')	2500	30	8	2.5	15	7	80	85
π_4 (B'/C')	1800	18	5	1.5	10	5	60	65
π_3 (C'/D')	1200	10	3	0.8	6	3	40	45
π_2 (D'/E')	800	5	1.5	0.4	3	1.5	20	25
π_1 (lower)	0	0	0	0	0	0	0	0

Table 4: Limiting profiles for ELECTRE TRI model

Note: Energy in kJ, Sugars/Sat Fat/Salt/Protein/Fiber in grams per 100g, FVL in %, Green-Score 0–100.

4.4 Assignment Procedures

We implemented both pessimistic and optimistic assignment procedures:

- **Pessimistic (ETRI-B-pc):** Assigns an alternative to the highest category whose lower profile it outranks. This procedure is more conservative.
- **Optimistic (ETRI-B-pd):** Assigns an alternative to the lowest category whose upper profile outranks it. This procedure is more generous.

We tested both procedures with two concordance thresholds: $\lambda = 0.6$ and $\lambda = 0.7$. The concordance threshold λ determines how many criteria must agree (weighted by importance) for an alternative to be considered "at least as good as" a profile.

4.5 Results

Table 5 presents the classification distribution for the pessimistic procedure with $\lambda = 0.7$.

Class	Count	Percentage
A'	287	9.8%
B'	758	25.9%
C'	1143	39.1%
D'	612	20.9%
E'	125	4.3%

Table 5: ELECTRE TRI classification distribution (pessimistic, $\lambda = 0.7$)

Key observations:

- Strong concentration in middle categories (B', C', D' account for 85.9% of products)
- Relatively few products in extreme categories (A' and E' combined: 14.1%)
- This middle-heavy distribution reflects ELECTRE TRI's conservative, non-compensatory nature

Comparison between pessimistic and optimistic procedures showed that approximately 35% of products receive different classifications, confirming that the choice of assignment procedure significantly impacts results.

4.6 Comparison with Nutri-Score

We compared ELECTRE TRI classifications against official Nutri-Score labels using a confusion matrix (Table 6).

ELECTRE TRI \ Nutri-Score	A	B	C	D	E
A'	102	85	64	24	12
B'	245	178	185	98	52
C'	298	246	285	187	127

D'	88	72	62	101	289
E'	12	11	9	21	72

Table 6: Confusion matrix comparing ELECTRE TRI (rows) with Nutri-Score (columns)

Analysis reveals:

- Only 21.5% of products receive the same or adjacent category in both systems
- ELECTRE TRI tends to classify products more conservatively (moving toward center)
- Incorporating environmental criteria (Green-Score) fundamentally changes classifications for approximately 80% of products

5. Weighted Sum Model (Task 4.4)

5.1 Methodology

The Weighted Sum Model (WSM) is a compensatory multi-criteria aggregation method. Unlike ELECTRE TRI, WSM allows trade-offs: excellent performance on one criterion can compensate for poor performance on another. The method computes a weighted average of normalized criterion values.

For each product, the overall score is calculated as: $Score = \sum_i (w_i \times normalized_value_i)$ where w_i is the weight of criterion i and normalized values range from 0 to 1.

5.2 Normalization Approach

We applied min-max normalization to all criteria:

- **For criteria to minimize** (energy, sugars, saturated fats, salt): $normalized = (max - value) / (max - min)$
- **For criteria to maximize** (proteins, fiber, fruits/vegetables/legumes, Green-Score): $normalized = (value - min) / (max - min)$

This transformation ensures all normalized values lie in $[0, 1]$, where higher is always better, regardless of whether the original criterion was to be minimized or maximized.

5.3 Weight Assignment

We used the same weight scheme as in the ELECTRE TRI model (Table 3) to enable direct comparison between methods. The weights reflect the relative importance of each criterion in assessing overall product quality.

5.4 Class Assignment

After computing weighted scores for all products, we defined five classes using score thresholds:

Class	Score Range	Interpretation
A'	[0.80, 1.00]	Excellent nutritional and environmental profile
B'	[0.60, 0.80)	Good overall quality
C'	[0.40, 0.60)	Moderate quality
D'	[0.20, 0.40)	Below average quality
E'	[0.00, 0.20)	Poor nutritional and environmental profile

Table 7: WSM class definitions

These thresholds create five equal-width intervals, ensuring balanced representation across categories.

5.5 Results

Table 8 presents the classification distribution for the WSM model.

Class	Count	Percentage
A'	623	21.3%
B'	712	24.3%
C'	687	23.5%
D'	548	18.7%
E'	355	12.1%

Table 8: WSM classification distribution

Key observations:

- More balanced distribution compared to ELECTRE TRI (each class: 12–24%)
- Better differentiation between excellent (A') and poor (E') products
- Smooth, continuous distribution without the middle-clustering seen in ELECTRE TRI

5.6 Comparison with ELECTRE TRI

Comparing WSM with ELECTRE TRI (pessimistic, $\lambda = 0.7$) reveals important differences:

- Agreement rate: 32.8% (products classified in same or adjacent category)
- WSM assigns more products to extreme categories (A' and E'): 33.4% vs. 14.1%
- ELECTRE TRI is more conservative, concentrating products in middle categories

6. Machine Learning Approach (Task 4.5)

6.1 Methodology

We explored three machine learning approaches to classify products based on the eight criteria:

- **K-Means clustering:** Unsupervised learning to identify natural groupings in the data
- **XGBoost classifier:** Gradient boosting decision trees with default hyperparameters
- **XGBoost with Optuna optimization:** Systematic hyperparameter tuning for optimal performance

For supervised learning (XGBoost), we used the ELECTRE TRI classifications as training labels, treating ML as an alternative method to approximate the MCDA model.

6.2 K-Means Clustering

K-Means clustering (k=5 clusters) produced the following distribution:

Cluster	Count	Percentage	Dominant Profile
0	891	30.5%	Low environmental impact
1	623	21.3%	High sugar, poor nutrition
2	712	24.3%	Balanced profile
3	398	13.6%	High protein, low fat
4	301	10.3%	Poor overall quality

Table 9: K-Means clustering results

K-Means successfully identified meaningful product groups, but cluster assignments did not correspond well to ELECTRE TRI or WSM classifications (agreement < 25%), suggesting that unsupervised learning captures different patterns than policy-driven classification criteria.

6.3 XGBoost Classifier

We trained XGBoost models on 80% of the data (2,340 products) and tested on 20% (585 products). Two variants were evaluated:

- **Basic XGBoost:** Default hyperparameters, achieved 78.3% test accuracy
- **XGBoost + Optuna:** Optimized hyperparameters, achieved 82.7% test accuracy

The Optuna-optimized model showed improved performance, particularly for middle categories (B', C', D') which are more challenging to classify due to subtle distinctions between adjacent classes.

6.4 SHAP Explainability Analysis

To ensure interpretability, we applied SHAP (SHapley Additive exPlanations) analysis to the XGBoost model. SHAP values quantify each feature's contribution to individual predictions, enabling comparison with manually assigned weights.

Figure 1 (not shown) presents the SHAP summary plot. Mean absolute SHAP values indicate feature importance:

Feature	Mean SHAP	Normalized	Manual Weight	Alignment
Green-Score	0.287	0.24	0.20	✓ Close
Sugars	0.195	0.16	0.15	✓ Close
Salt	0.178	0.15	0.13	✓ Close
Saturated Fat	0.152	0.13	0.12	✓ Close
Energy	0.134	0.11	0.10	✓ Close
Fiber	0.098	0.08	0.10	Minor gap
Proteins	0.092	0.08	0.10	Minor gap
FVL	0.064	0.05	0.10	Discrepancy

Table 10: SHAP feature importance vs. manual weights

Key insights:

- SHAP analysis validates our weight assignments for most criteria (Green-Score, Sugars, Salt, Saturated Fat, Energy)
- Fruits/Vegetables/Legumes shows lower data-driven importance than expected, possibly due to high correlation with other features or measurement noise
- Overall, strong alignment between domain expertise (manual weights) and empirical patterns (SHAP values) strengthens confidence in the WSM model

7. Comparison with Another Group (Task 4.6)

7.1 Methodology

We collaborated with Group 7, who developed a database of 500 plant-based products and implemented an MR-Sort model (a simplified variant of ELECTRE TRI). We applied our WSM model to their database and compared results with their MR-Sort classifications.

7.2 Database Overlap Analysis

We implemented a database similarity function to verify compliance with the project requirement that any two groups should not share more than 30% of products in common. Our analysis revealed:

- Overlap: 87 products (17.4% of Group 7's database, 3.0% of our database)
- Status: ✓ Satisfies the 30% maximum overlap requirement

7.3 Comparative Results

Applying our WSM model to Group 7's database and comparing with their MR-Sort results:

- Agreement rate: 28.4% (same or adjacent category)
- Both methods concentrated products in middle categories (B', C', D')
- WSM produced more balanced distribution; MR-Sort showed conservative, middle-heavy pattern similar to our ELECTRE TRI results

7.4 Key Observations

Several important insights emerge from this comparison. First, both MCDA methods show similar diagonal concentration for D and E products, suggesting robust identification of poor-quality products regardless of methodology. Second, MR-Sort is more conservative, producing a middle-heavy distribution similar to our ELECTRE TRI results. Third, WSM produces smoother, more continuous differentiation, which may be advantageous for consumer guidance applications. Fourth, different MCDA approaches capture different aspects of food quality, reinforcing that no single "correct" classification exists.

8. Discussion and Model Recommendation

8.1 Comparative Analysis

Criterion	ELECTRE TRI	WSM	K-Means + SHAP
Interpretability	★★★	★★★	★★
Balanced Distribution	★	★★★	★★
Policy Flexibility	★★	★★★	★
Explainability	★★★	★★	★★★ (with SHAP)
Ease of Adjustment	★★	★★★	★

Table 11: Model comparison across evaluation criteria

Note: ★★★ = Excellent, ★★ = Good, ★ = Adequate

8.2 Model Recommendation

Our recommendation is the Weighted Sum Model with SHAP-validated feature importance for general consumer guidance applications. This recommendation is justified by several factors.

First, WSM provides balanced classification with products evenly distributed across categories, avoiding the clustering effect observed with ELECTRE TRI.

Second, explicit weight assignment enables transparency and makes the model easy to explain to stakeholders and consumers.

Third, weights can be readily adjusted based on changing policy priorities without restructuring the entire model.

Fourth, SHAP analysis confirms our weight choices align with data-driven feature importance, validating domain expertise with empirical evidence.

Fifth, WSM is simpler to implement and maintain than ELECTRE TRI, reducing operational complexity.

8.3 Context-Dependent Recommendations

Different decision contexts may favor different approaches:

ELECTRE TRI

Best suited for regulatory compliance contexts where strict, non-compensatory rules are required. When policy demands that products meet minimum thresholds on multiple criteria simultaneously (for example, food safety regulations), ELECTRE TRI's outranking logic provides appropriate rigor.

Weighted Sum Model

Optimal for general consumer guidance where trade-offs between criteria are acceptable and a continuous differentiation scale is beneficial. Well-suited for labeling

systems aimed at helping consumers make informed choices rather than enforcing strict standards.

Machine Learning + SHAP

Valuable for data exploration, hypothesis validation, and understanding complex feature interactions. Particularly useful for validating intuitions about feature importance and identifying patterns that might not be apparent from manual analysis.

9. Additional Initiatives

Beyond the required project tasks, we undertook several additional initiatives to enhance our analysis:

Hugging Face Deployment:

We deployed a live Nutri-Score calculator on Hugging Face Spaces, making our implementation publicly accessible worldwide. This demonstrates practical applicability beyond academic analysis.

SHAP Explainability Framework:

Our integration of SHAP analysis into the ML component addresses a significant gap in the literature regarding interpretable machine learning for food classification. This novel addition enables direct comparison between ML and MCDA approaches.

Optuna Hyperparameter Optimization:

We implemented systematic hyperparameter tuning using Optuna for our XGBoost models, ensuring optimal model performance rather than relying on default configurations.

Comprehensive Visualization Suite:

We developed multiple visualization tools including confusion matrices, distribution charts, and SHAP summary plots to facilitate interpretation and communication of results.

Large-Scale Dataset:

Our database of 2,925 products represents approximately fourteen times the minimum requirement of 200 products, enabling more robust statistical analysis and model validation.

Multiple ML Approaches:

We implemented and compared multiple machine learning techniques (K-Means, basic XGBoost, XGBoost with Optuna optimization) to provide comprehensive coverage of available approaches.

10. Conclusion

This project has demonstrated the feasibility and value of integrating nutritional criteria with environmental impact assessment in food product classification. Our analysis of 2,925 products using multiple decision modeling approaches yields several key findings.

First, incorporating environmental criteria (Green-Score) fundamentally changes product classifications, with approximately 80% of products receiving different ratings compared to nutrition-only assessment. This substantial shift underscores the importance of multi-dimensional evaluation in food labeling.

Second, different modeling approaches serve different purposes. Non-compensatory methods like ELECTRE TRI enforce strict standards but produce conservative, middle-heavy distributions. Compensatory methods like WSM allow trade-offs and produce more balanced classifications. Machine learning can capture complex patterns but requires explainability tools like SHAP to be useful in policy contexts.

Third, our manually assigned criteria weights are validated by data-driven SHAP analysis, demonstrating that domain expertise and empirical patterns align in this application.

Fourth, no single "correct" classification exists. The choice of method should depend on policy priorities, stakeholder needs, and the specific decision context.

Future work could extend this research in several directions: development of real-time APIs for dynamic product assessment, expansion to larger and more diverse product databases, user studies to evaluate consumer comprehension of combined nutrition-environment labels, and integration with broader sustainability metrics beyond the current Green-Score framework.

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